Mohammed Faizan

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Education

PES University — Bengaluru, KA
B.Tech in Computer Science (Artificial Intelligence & Machine Learning) — *2024 – Present*

Shaheen PU College — Bengaluru, KA
State Board of Karnataka – 95.88% — *2022 – 2024*

GEM School — Bengaluru, KA
State Board of Karnataka – Grade 10, 97% — *2021 – 2022*

Technical Skills

Languages: Python, C/C++, JavaScript, HTML/CSS, Verilog, Rust
Developer Tools: Git, Docker, VS Code, Cursor, Linux, Kali Linux, Kubernetes, Netlify, Wazuh
Libraries: Scikit-learn, BeautifulSoup, NumPy, Matplotlib, TensorFlow, OpenCV, SciPy, Hugging Face
Web/AI Frameworks: React, Node.js, Next.js, Tailwind CSS, FastAPI, LangChain, LangGraph
Databases: SQL, PostgreSQL, MongoDB, Mongo Express, Neo4j, Redis, ChromaDB

Experience

Research Intern — Jun 2026 – Present
Center for Cloud Computing and Big Data, PES University — *Bengaluru, India*

- Engineered TRISP, a real-time safety-aware routing engine for Bengaluru's 65 km^2 street network using live TomTom and OpenWeatherMap APIs.
- Designed TSA-NAMOA*, a twin-probe algorithm resolving Pareto Front Collapse in NAMOA* under peak-hour traffic, backed by a 3-stage Mistral-7B RSS crime-intelligence pipeline.
- Beat 5-factor literature baselines in a 600-query evaluation: 0% worst-case failures, +2.28% cost reduction, 1.03s latency; co-authored an IEEE submission.

Junior Manager, Incoming Corporate Sector — Aug 2025 – Jul 2026
AIESEC in Bengaluru — *Bengaluru, KA*

- Led cold-calling outreach and market research to secure client meetings with Bengaluru startups.
- Managed global AIESEC relations, matching international applicants with corporate internships across India.

Teaching Assistant — Feb 2026 – Present
PES University — *Bengaluru, KA*

- **Python Programming** (Sep 2026 – Present): Design lecture slides and HackerRank coding challenges; evaluate student submissions for code quality and problem-solving.
- **C Programming** (Feb 2026 – May 2026): Taught programming fundamentals and designed quizzes/study materials to reinforce core concepts.

Projects

Agentic AI SOC Analyst | *Python, LangGraph, Wazuh SIEM, PostgreSQL, ChromaDB, HTMX*

- Built a team of AI agents using LangGraph and Wazuh SIEM that automatically review security alerts and investigate threats.
- Secured the AI pipeline against prompt injection with a layered defense system and a live dashboard blocking malicious IPs.

BharatKey – Open-Source Hardware Security Key | *C, STM32, USB, mbedTLS*

- Built a FIDO2/WebAuthn hardware security key on an STM32F411 MCU for passwordless authentication (GitHub, Google).
- Implemented CTAP2 authentication and smart-card PDF signing via ECDSA P-256 with mbedTLS and secure flash storage.

Multi-Material Defect Inspection System (Edge AI) | *PyTorch, YOLOv10, FastAPI, ESP32*

- Architected an edge computer vision pipeline (ResNet-18 + YOLOv10) detecting 25 defect classes across Steel, Aluminum, and Wood, boosting micro-scratch recall by +14.2% via a custom shape-descriptor module.
- Built a low-latency FastAPI/WebSocket backend streaming live diagnostics at ˜45ms/frame (22 FPS), integrated with ESP32 to trigger real-time defect-rejection hardware.

Extracurricular Activities

Head of Volunteers & Student Coordinator

AI 360˚ Unmasking AI for Society – Pre-Summit Event of India-AI Impact Summit 2026, PES University

- Led volunteer operations and cross-team design/marketing/finance coordination for the Pre-Summit event.

Campus Involvement: Member of AIESEC, Samarpana, Weal, DEBSOC, and AL Coding

Achievements

- 1st Place – Ignite Room (Bangalore Zonal level); 3rd Place – National-Level Cybersecurity Hackathon.
- Top 6 – Computer Vision National-Level Hackathon (organized by RVCE); Top 10 in 4 hackathons, Top 15 in 2 CTFs.
- Mentor – State-Level Hackathon; 4x CNR Scholarship recipient; Best Outgoing Student of the Year – 2022.